

山东大学 计算机科学与技术 学院

大数据分析实践 课程实验报告

学号：202300130041	姓名：徐守政（小组任务）	班级：数据
实验题目：BERT 环境配置		
实验学时：2	实验日期：2025/10/31	
实验目的：能够在服务器上配置 BERT 环境		
硬件环境： 计算机一台		
软件环境： Linux，AutoDL 算力云		

实验步骤与内容：

服务器的租借

实例ID / 名称	状态	规格详情	本地磁盘	健康状态	付费方式	释放时间/停机时间	SSH登录	快捷工具	操作
北京B区 / 260机 1b2041a5c3-1e506ba2 设置名称	● 关机中	RTX 4090 * 1卡 查看详情	系统盘 35.89% 数据盘 0.00%	● 正常	按量计费 余额不足24小时	14天23小时59分后释放 设置定时关机			开机 更多

服务器 ssh 连接

```
PS D:\Desktop\data> ssh -p 45194 root@connect.bjb2.seetacloud.com
The authenticity of host '[connect.bjb2.seetacloud.com]:45194 ([106.38.204.136]:45194)' can't be established.
ED25519 key fingerprint is SHA256:liZ36vNCsNcNdXeWs4f+g5ZiHPM/ZihP834vxs8U1qc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[connect.bjb2.seetacloud.com]:45194' (ED25519) to the list of known hosts.
root@connect.bjb2.seetacloud.com's password:
Permission denied, please try again.
root@connect.bjb2.seetacloud.com's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-119-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

//root/autodl-tmp || 数据盘 || 快 || 实例关机数据不会丢失，可存放读写IO要求高的数据。但不会随保存镜像一起保存 ||

CPU   : 25 核心
内存  : 90 GB
GPU   : NVIDIA GeForce RTX 5090, 1
```

虚拟环境的创建

```
root@autodl-container-0b944c8bb0-7425bfb8:~# conda create -n bert-mrpc python=3.8
Channels:
 - https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
 - https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/free
 - https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud/pytorch
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: /root/miniconda3/envs/bert-mrpc

  added / updated specs:
    - python=3.8

The following packages will be downloaded:
```

展示虚拟环境

```
root@autodl-container-0b944c8bb0-7425bfb8:~# conda env list
# conda environments:
#
base                                /root/miniconda3
bert-mrpc                          /root/miniconda3/envs/bert-mrpc

root@autodl-container-0b944c8bb0-7425bfb8:~# █
```

激活虚拟环境

```
(base) root@autodl-container-0b944c8bb0-7425bfb8:~# conda activate bert-mrpc
(bert-mrpc) root@autodl-container-0b944c8bb0-7425bfb8:~# █
```

安装 torch 库

```
(base) root@autodl-container-0b944c8bb0-7425bfb8:~# conda activate bert-mrpc
(bert-mrpc) root@autodl-container-0b944c8bb0-7425bfb8:~# conda activate bert-mrpc
pip install torch==1.7.0 transformers==4.18.0
pip install numpy pandas tqdm
Looking in indexes: http://mirrors.aliyun.com/pypi/simple
Collecting torch==1.7.0
  Downloading http://mirrors.aliyun.com/pypi/packages/05/2d/bb4611cf053eaa679f6086b78cff2776ff1d51a15fe5e063cdcbfc6b5577/torch-1.7.0-cp38-cp38-manylinux1\_x86\_64.whl (776.8 MB)
    776.8/776.8 MB 983.7 kB/s eta 0:00:00
Collecting transformers==4.18.0
  Downloading http://mirrors.aliyun.com/pypi/packages/8f/e9/c2b4c823b3959d475a570c1bd2df4125478e2e37b96fb967a87933ae7134/transformers-4.18.0-py3-none-any.whl (4.0 MB)
    4.0/4.0 MB 1.1 MB/s eta 0:00:00
Collecting future (from torch==1.7.0)
  Downloading http://mirrors.aliyun.com/pypi/packages/da/71/ae30dadffc90b9006d77af76b393cb9dfbfc9629f339fc1574a1c52e6806/future-1.0.0-py3-none-any.whl (491 kB)
Collecting typing-extensions (from torch==1.7.0)
  Downloading http://mirrors.aliyun.com/pypi/packages/0b/54/b1ae86c0973cc6f0210b53d508ca3641fb6d0c56823f288d108bc7ab3cc8/typing\_extensions-4.1.2-py3-none-any.whl (45 kB)
Collecting dataclasses (from torch==1.7.0)
  Downloading http://mirrors.aliyun.com/pypi/packages/26/2f/1095cdc2868052dd1e64520f7c0d5c8c550ad297e944e641dbf1ffbb9a5d/dataclasses-0.6-py3-none-any.whl (14 kB)
Collecting numpy (from torch==1.7.0)
```

在 pycharm 上配置服务器链接



下载 BERT 模型的时候无法在 Hugging Face 官方站下载，采用镜像站下载

```

用Wget从镜像站下载4个文件 (AutoDL支持断点续传)
wget https://hf-mirror.com/bert-base-uncased/resolve/main/config.json
wget https://hf-mirror.com/bert-base-uncased/resolve/main/pytorch_model.bin
wget https://hf-mirror.com/bert-base-uncased/resolve/main/tokenizer_config.json
wget https://hf-mirror.com/bert-base-uncased/resolve/main/vocab.txt
2025-10-31 20:32:32 https://hf-mirror.com/bert-base-uncased/resolve/main/config.json
Connecting to 172.32.52.144:12798... connected.
Proxy request sent, awaiting response... 200 OK
Length: 570 [text/plain]
Saving to: config.json.2'

config.json.2 100%[=====] 570 --KB/s in 0s

2025-10-31 20:32:34 (28.5 MB/s) - config.json.2' saved [570/570]

-2025-10-31 20:32:34- https://hf-mirror.com/bert-base-uncased/resolve/main/pytorch_model.bin
Connecting to 172.32.52.144:12798... connected.
Proxy request sent, awaiting response... 200 OK
Length: 420.07M [application/octet-stream]
Saving to: pytorch_model.bin.2

pytorch_model.bin.2 100%[=====] 420.07M 20.8MB/s in 19s

2025-10-31 20:32:55 (21.7 MB/s) - pytorch_model.bin.2' saved [440473133/440473133]

-2025-10-31 20:32:55- https://hf-mirror.com/bert-base-uncased/resolve/main/tokenizer_config.json
Connecting to 172.32.52.144:12798... connected.
Proxy request sent, awaiting response... 200 OK
Length: 48 [text/plain]
Saving to: tokenizer_config.json.2'

tokenizer_config.json.2 100%[=====] 48 --KB/s in 0s

2025-10-31 20:32:55 (3.36 MB/s) - tokenizer_config.json.2' saved [48/48]

-2025-10-31 20:32:55- https://hf-mirror.com/bert-base-uncased/resolve/main/vocab.txt
Connecting to 172.32.52.144:12798... connected.
Proxy request sent, awaiting response... 200 OK
Length: 231508 (226K) [text/plain]
Saving to: vocab.txt.2'

```

实例训练可行，证明 BERT 环境配置成功

结论分析与体会：

配置 BERT 环境实验的核心结论是，成功搭建需围绕“依赖版本兼容、硬件驱动适配、环境隔离规范”三大关键要素：Python 需控制在 3.8-3.9 版本以适配 Transformers 4.18.0-4.25.0 主流稳定版，同时 PyTorch/TensorFlow 版本需与 CUDA 版本严格对应（如 PyTorch 1.10.0 匹配 CUDA 11.3），避免因版本不兼容导致的库函数报错或 GPU 调用失败；硬件层面需确保 NVIDIA 显卡驱动正常安装（可通过 nvidia-smi 验证），否则会出现 GPU 无法识别、显存分配失败等问题，而使用 Anaconda 或虚拟环境（如实验中的 bert-mrpc 环境）能有效隔离不同项目的依赖冲突，降低环境污染风险。从实验体会来看，配置过程中需耐心排查报错信息（如“command not found”多为驱动或路径问题，“version mismatch”为依赖版本冲突），优先参考官方文档的环境要求而非盲目使用最新版本，同时提前验证硬件支持情况（如老旧显卡可能不兼容高版本 CUDA），这些细节能大幅减少调试时间，也让我意识到深度学习环境配置并非单纯的“安装软件”，而是兼顾兼容性、稳定性与硬件适配的系统性工作，为后续模型训练奠定基础。